

FlixPulse

SIDDHARTHA GOSWAMI
23323036

Latent Factor Movie Recommendation & Analytics

Personalized Movie Discovery Engine using the Netflix Prize Dataset

21K+ Users

Active Ratings

1,287 Movies

Dense Subset

5.7M Ratings

Training Data

0.816 RMSE

SVD Accuracy

The Personalization Challenge

Sparsity, Latency & Cold-Start



Data Sparsity

Real-world user logs are extremely sparse — over 99% of ratings are missing, making reliable inference difficult.



Inference Latency

Production recommendation services must resolve personalized results within milliseconds at scale.



Cold-Start Problem

New users and movies have zero historical ratings, leaving standard collaborative filters unable to predict.



Explainability Gap

ML predictions are often 'black boxes' — users need transparent reasoning to trust recommendations.

Netflix Subset Insights

& User Statistics

21,045

Active Users

60%+

Ratings are 4★ or 5★

Mean rating 3.73 — strong positive selection bias in the dataset.

1,287

Movies

21%

Matrix Density

78.96% of the rating matrix is sparse — a core challenge for the models.

5.7M

Ratings

Long Tail Distribution

Blockbuster titles capture a disproportionate share of all reviews.

SVD Latent Factors

& KNN Neighborhood Models

Matrix Factorization (SVD)

Learns latent user vectors p_u and item vectors q_i from rating data

Predicts: $\hat{r}_{u,i} = \mu + b_u + b_i + p_u^T q_i$

Bias terms (b_u, b_i) capture systematic rating tendencies

Optimized with Stochastic Gradient Descent (SGD)

VS

Item-Based KNN Model

Computes item similarities using adjusted Cosine Correlation

Predicts via weighted average of nearest-neighbor rated items

Naturally interpretable — explains why items are related

No latent factors; similarity is directly observable

EVALUATION RESULTS

RMSE Prediction

vs. MAP@10 Ranking Performance

WINNER

0.8160

SVD · RMSE

Rating accuracy

WINNER

0.7427

SVD · MAP@10

Top-10 ranking quality

0.8553

KNN · RMSE

Rating accuracy

0.7079

KNN · MAP@10

Top-10 ranking quality

Training Time Comparison

SVD
8.9s

KNN 225.9s (25× slower)

Combining SVD Accuracy with KNN Interpretability

The Trade-off

SVD is highly accurate but uninterpretable — users receive no reason for recommendations.

KNN is naturally explainable, but less accurate and significantly slower to train.



Hybrid Solution

SVD generates the scored candidates and final ranking list.

KNN similarity weights are applied post-hoc to produce human-readable explanations for each result.



Example Output



*"Because you rated
Toy Story 5 ★
(Similarity: 98%),
we recommend
Toy Story 2
(SVD score: 4.8★)"*

Handling Cold-Start

for Users and Movies



New User

Fall back to popularity recommender

Return movies with highest mean rating

Threshold: $\geq 2,000$ existing ratings



New Movie

Genre metadata overlap for similarity

ϵ -greedy exploration injects new releases

Exposure to active users guaranteed



Sparse History

Mix popularity averages + hybrid predictions

Item-based KNN fills gaps from partial history

Confidence-weighted blending strategy

Decoupled Web Application & Interactive Dashboard

Backend

FastAPI REST server with cached recommendations, similarity lookups, and sub-50ms p99 latency. Modular endpoints for personalized feeds, item lookup, and explanation retrieval.

Frontend

Vite-based React SPA with premium dark-mode glassmorphism styling, custom SVG stats charts, and animated discovery feed. Fully responsive and production-ready.

Future Scope

-  Deep Neural Collaborative Filtering networks for higher-capacity latent representations
-  TMDB scraper integration for movie posters, trailers, and cast metadata
-  Real-time streaming feedback with session-aware contextual bandits
-  A/B testing framework for live ranking model comparison